

Roll no.

MID SEMESTER EXAMINATION SEP 2025

Name of the Program: B. Tech, ECE (ESR)

Semester: V

Name of the Course: Sensor Technology

Course Code: TEC 551

Time: 90mins

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
a)	What is hysteresis error? What is nonlinearity in sensors? Explain how both affect sensor performance.	CO1
	OR	
b)	Explain the difference between accuracy and precision in measurement, with examples.	
Q2	(10 marks)	
a)	Define span and full-scale output. A sensor has an input range of 0–200 N and an output range of 0–5 V. (i) What is its span? (ii) What is its full-scale output?	CO1
	OR	
b)	What do you mean by the transfer function of a sensor? Why is it important? Illustrate with an example.	
Q3	(10 marks)	
a)	What are the DYNAMIC MODEL of sensor elements?	CO2
	OR	
b)	Differentiate between the piezoelectric effect and the pyroelectric effect, giving suitable sensor examples.	
Q4	(10 marks)	
a)	Explain the thermoelectric effect. How is it used in thermocouples for temperature measurement?	CO2
	OR	
b)	Define capacitance. Explain how a parallel-plate capacitor can be used as a displacement sensor.	
Q5	(10 marks)	
a)	What do you understand by repeatability, resolution, and dead band in sensors?	CO1 / CO 2
	OR	
b)	State and explain Faraday's law of electromagnetic induction. How is it applied in sensors?	

